

Cross-Sectional Equity Alpha: From Data Inspection to Backtest

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Abstract

This project builds a tradable, dollar-neutral cross-sectional equity alpha from a panel of security reference data, eleven signals (**d1** to **d11**) and six risk factors (**rf1** to **rf6**) spanning 2010 to 2017. The study runs from: data inspection and cleaning, point-in-time feature and target construction, model tuning and selection, and a walk-forward backtest. Every step is built to avoid look-ahead bias. The final chosen model is an ensemble forecast traded as a risk-factor-neutral, dollar-neutral portfolio. It delivers a net Sharpe ratio of 1.41 over the full window (2011 to 2017) at 3.4% annualized volatility.

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1 Introduction

The goal is to turn the provided cross-sectional data into a tradable alpha and to evaluate it strictly point-in-time. For each security i and trading day t the model forecasts a forward return close-to-close from $t + 1$ to $t + 2$, rank securities cross-sectionally, and hold a dollar-neutral portfolio rebalanced under realistic trading costs. I measure performance by the net (after transaction cost) Sharpe ratio of the walk-forward portfolio.

This project rigorously eliminates look-ahead bias. Every transformation or calculation on day t uses information available at or before t only. Cross-sectional normalization is computed within a single day, the target is a strictly forward return, and models are trained and validated on rolling windows. During evaluation, such

as feature selection, hyper-parameter tuning, or normalization statistics, the out-of-sample consequence is reported.

The features used in the model are built from `d1` to `d11` and the `group_id` of the security only. The six risk factors, `rf1` to `rf6` were not as alpha but to remove factor exposure from the final book. The work is organized in three notebooks: data inspection (`data_inspection.ipynb`), modeling (`alpha_model.ipynb`), and portfolio construction (`portfolio_construction.ipynb`).

2 Data Inspection

This section documents the data-quality inspection and the signal diagnostics that later feature engineering depends on. The major procedures and findings are documented below, please find more experimental results and detailed plots in `data_inspection.ipynb`.

2.1 Data quality and cleaning

- **Trading universe.** Only about 45% of rows are eligible to trade (`in_trading_universe` equal to Y: 4.00M of 8.8M), and the universe size changes slightly over time, including a visible drop in 2011 to 2012. Since the number of dropped securities is still relatively small, I've decided to still include them in the training data panel. The model is trained using only tradable securities, to keep the the model aligned with realizable positions and avoids learning from securities we could never take a position in.
- **Return outliers.** The daily return `ret1d` is fat-tailed and some values might be contaminated: the maximum is 1099 (that is, 109,900%), which are data errors rather than tradable moves, and 8,854 returns are missing. I therefore winsorized `ret1d` cross-sectionally at 1 and 99 percentiles each day before building the target or any return-based statistic.
- **Forward-return target.** The target is the forward return $\text{ret1d_fwd}(i, t) = \text{ret1d}(i, t + 2)$. Given that signals available after market close on t , and returns are calculated close to close from previous day, return on $t + 2$ is the earliest we can fully capture. Because a security can leave the universe and re-enter months later, a naive two-day shift would pair a row before the gap with a return after it and create a spurious forward return across the gap. I instead compute chunks of continuous trading days for each security_id, and shift only within each chunk (a gap of more than five trading days starts a new run), so no return is ever paired across a gap.
- **Duplicate keys.** Three signal files have duplicates with same (`data_date`, `security_id`) pair. `d3` and `d4` contain 127,266 duplicate key pairs that are exact copies (zero conflicts) and duplicates are simply dropped. `d9` has about 8.0 million duplicate-key pairs, of which 4.3 million conflict and are different in values, so I aggregate them it into two features, the per-key mean (`d9_mean`) and spread (`d9_spread`).
- **Redundant signal.** Based on correlation and distribution plots, `d11` is identified as exact `d10` timed by a uniformly random variable between 3 to 4. Therefore, `d10` and `d11` have rank correlation 1.0 and near-identical IC, so I dropped `d11` to avoid double counting.
- **Security groups.** Prices are strictly positive (one row has non-positive volume), and the `group_id` hierarchy resolves to 11 top-level groups (25, 72, and 176 at finer levels) with no outlier bucket, so I decided to use this top-level `group_id` as an input variable.
- **Nonlinearity.** For each signal, I calculate the per-quantile forward-return profiles, several signals are curved and non-monotone rather than clear linear patterns, with most of the returns concentrated in the extreme buckets and little in the middle, which indicates a nonlinear signal-to-return relationship and therefore justifies evaluating nonlinear models that can capture such threshold and saturation effects.

2.2 Signal forecasting power

As a proxy for signal forecasting power, I calculated the per-signal information coefficient, its annualized information ratio (IR), hit rate, and coverage. Table 1 and Figure 1 summarize it.

Table 1: Per-signal coverage and forecasting power, ranked by absolute IC.

Signal	Coverage	Missing	Mean IC	IC IR (ann.)	Hit rate
d3	1.5%	21.5%	+0.041	+3.69	75.4%
d1	1.5%	3.8%	+0.036	+3.57	75.1%
d5	1.5%	11.8%	+0.032	+2.76	77.5%
d2	1.5%	11.5%	+0.030	+2.64	80.3%
d6	1.5%	19.6%	−0.022	−2.05	78.0%
d9	92.1%	0%	−0.011	−1.72	61.5%
d7	1.5%	19.0%	−0.012	−1.21	77.6%
d4	1.5%	3.7%	+0.013	+1.08	76.4%
d10	56.2%	0%	+0.004	+0.69	55.6%
d11	56.2%	0%	+0.004	+0.68	55.7%
d8	97.6%	0%	−0.001	−0.13	49.6%

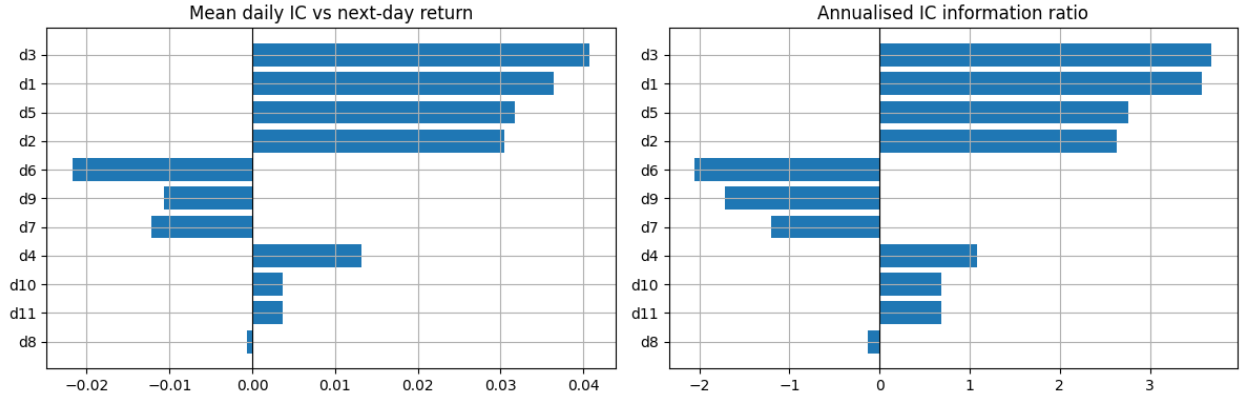


Figure 1: Signal IC strength and annualised IC information ratio

Two findings drive the modeling. First, forecasting power is concentrated. A handful of signals carry almost all of it: d3, d1, d5, and d2 (positive) and d6, d7, and d9 (negative) have information ratios between roughly 1.7 and 3.7 with hit rates of 75% to 80%, while d8 is essentially noise (IC near zero) and d10 and d11 are weak and redundant. Second, the strongest signals are the sparsest: d1 to d7 cover only about 1.5% of tradable name-days, whereas the high-coverage signals d8 and d9 are weak. Thus, for the most sparse signals d1 and d2, I decided to include two indicators for whether they are NaN as engineered features. This trade-off explains why a sparse, robust, IC-weighted combination later outperforms a all-variable regression. A signal is useful only if its IC is persistent rather than concentrated in one period, so I also examine the rolling IC over time in Figure 2.

2.3 Signal distributions and collinearity

The raw signals are heavy-tailed and mixed-sign (Figure 3), which rules out normal scaling or min-max scaling and motivates the rank-based normalization used in modeling. The collinearity structure (Figure 4) shows that several signals move together but that the information is not perfectly duplicated apart from the d10 and d11 pair noted above; a Pearson-versus-Spearman comparison further shows several pairs (d1&d2, d2&d3, and d4&d5) that look uncorrelated under raw Pearson yet are strongly monotone, since heavy tails break linear correlation, again favoring rank-based methods.

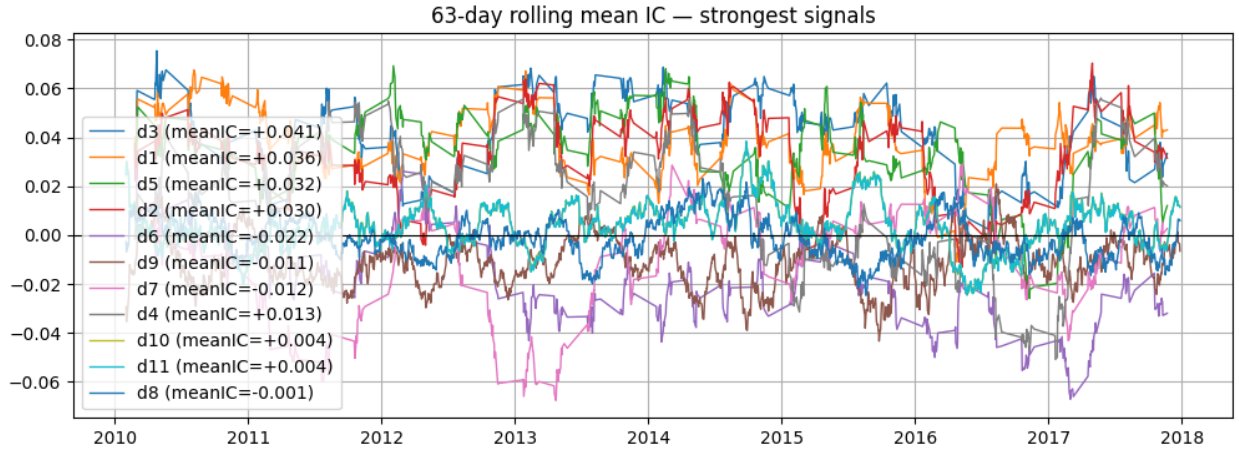


Figure 2: IC stability over time (rolling mean IC)

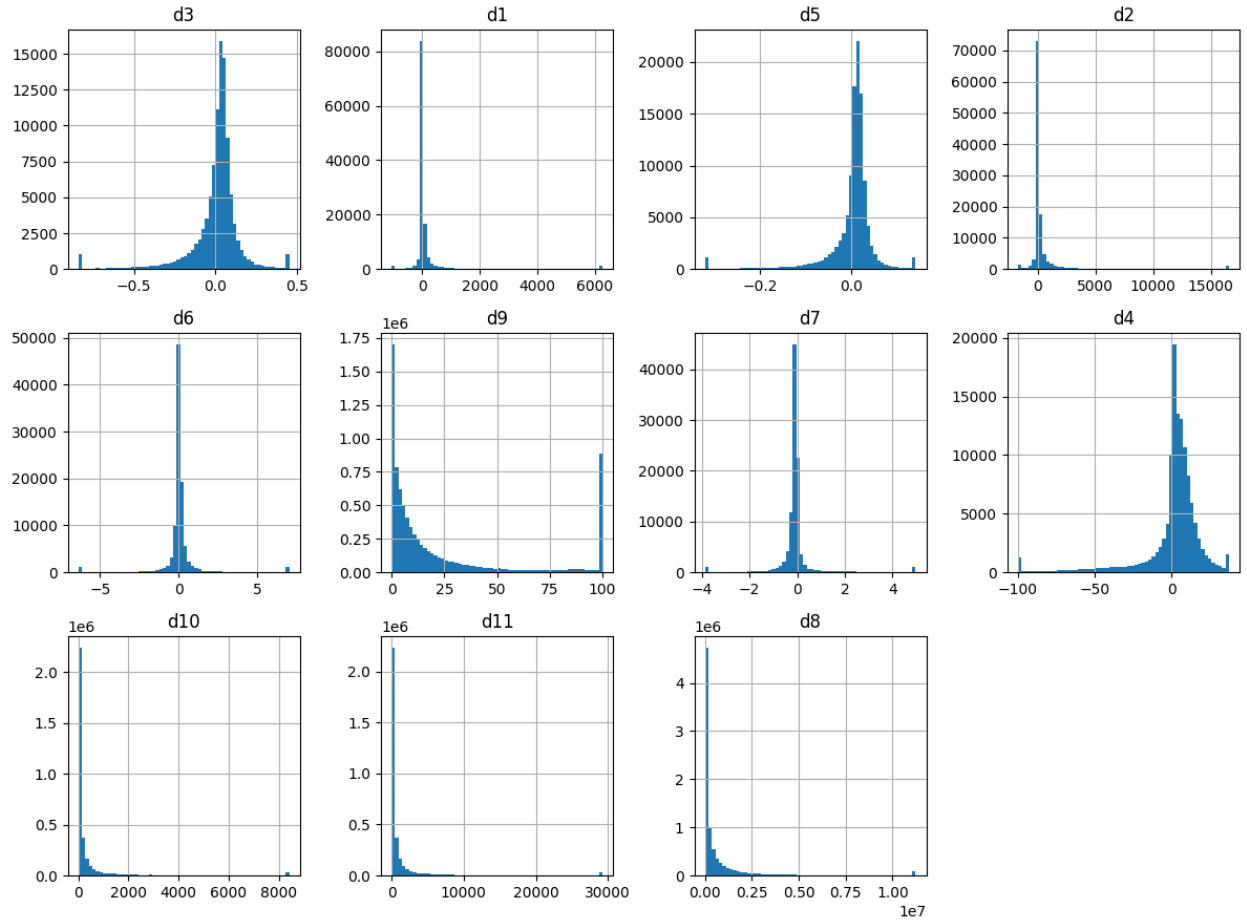


Figure 3: Signal distributions (robust-quantile clipped)

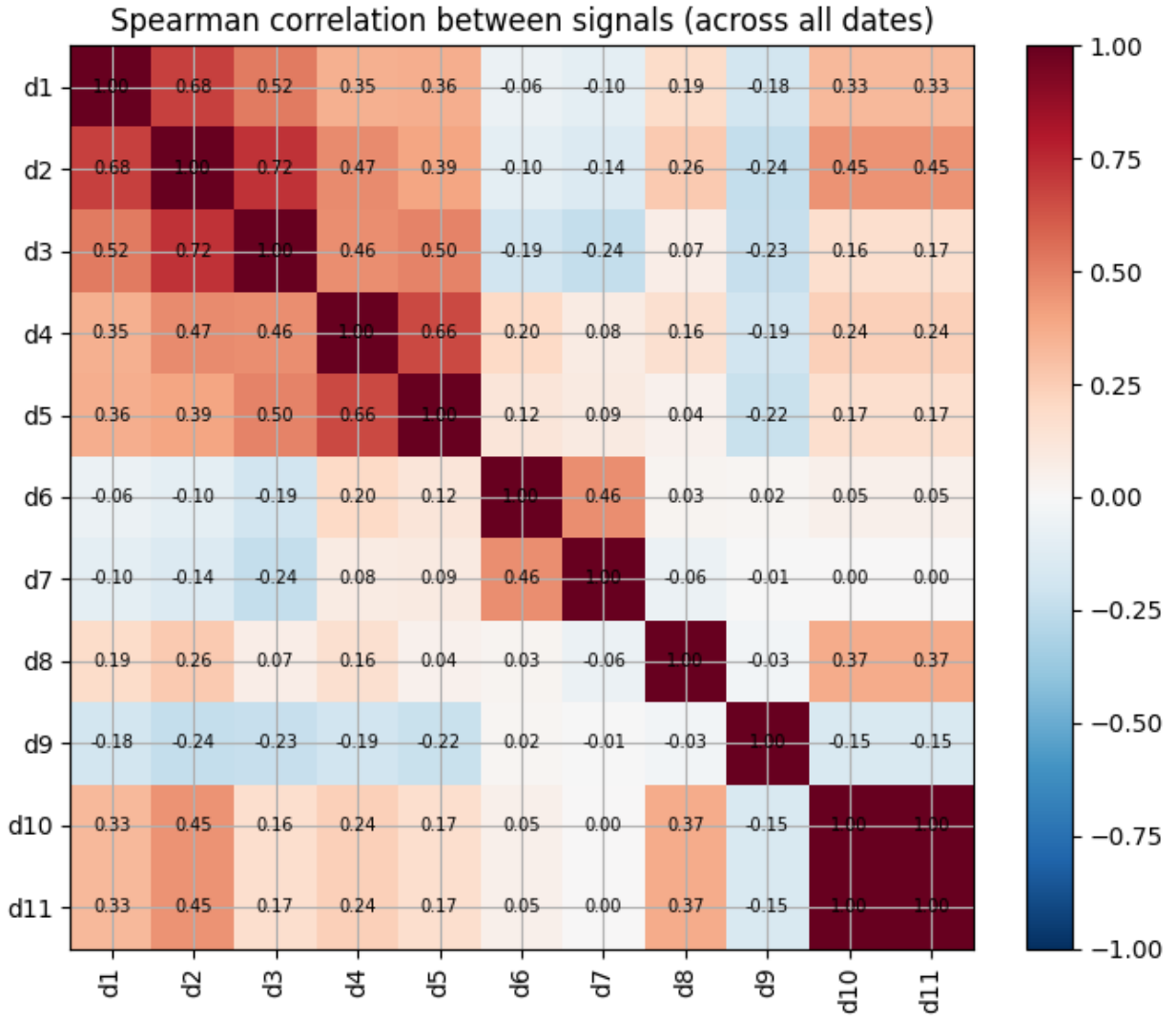


Figure 4: Signal collinearity (Spearman rank correlation)

2.4 Risk factors

I characterized the six risk factors and confirmed that the signals are not disguised factor exposures: no signal is strongly correlated with any of `rf1` to `rf6`. The alpha is therefore largely orthogonal to the factor structure, which justifies using the risk-factor neutralization in portfolio construction to reduce volatility without weakening signals.

3 Modeling

3.1 Target and feature construction

The prediction target is the winsorized forward return, demeaned cross-sectionally each day so that the model learns relative ranking rather than the market’s daily level. For features we apply one consistent transform to every signal and risk factor: a cross-sectional rank followed by an inverse-normal mapping, computed within each day. I chose this rank-Gaussian transform over min-max or z-scoring for three reasons that the data inspection made clear. It is robust to the extreme tails and data errors in the raw signals, it handles the mixed-sign sparse signals and the positive heavy-tailed ones on the same footing, and because it is purely cross-sectional it cannot leak information across time. After this transform, filling a missing value with zero correctly means “the cross-sectional median” rather than an extreme value.

3.2 Objective: information coefficient, not mean squared error

I select models on the information coefficient rather than on mean squared error or R^2 . I initially tested models by mean squared error and R^2 , but the fitted values were dominated by noise: the cross-sectional R^2 sat near 10^{-4} , the predictions were unstable from day to day, and the metric barely separated good models from bad, which is what made me switch to the information coefficient. Daily returns are almost unpredictable in level, so squared-error metrics barely separate models and are not aligned with what a cross-sectional book actually monetizes, namely the correct rank ordering of names each day. The IC measures exactly that rank-ordering skill and is the same quantity the portfolio converts into return, so optimizing it keeps model selection and trading objective consistent.

3.3 Model comparison and choice

I majorly compared four families on the same walk-forward protocol: forward-stepwise linear regression, principal-component regression, regularized linear models (ridge, lasso, and elastic net), and gradient-boosted trees (XGBoost and LightGBM). There are mainly two reasons behind this design. First, with only about a dozen weak, mostly linear signals, the gradient-boosted trees did not beat a well-regularized linear model out of sample and tended to add turnover without adding return, so the extra complexity was not justified. Second, no single linear model dominated. I therefore use an **ensemble** of two complementary point-in-time models:

1. an **IC-weighted composite**, which in each window weights each signal by its trailing information coefficient and sums them, giving a robust, low-variance, low-turnover forecast;
2. a **ridge regression** on the standardized signals together with the risk factors, refit each month, which captures joint linear structure the composite ignores.

The traded forecast is the average of the per-day cross-sectional z-scores of the two models. As Section 5 shows, this ensemble produces a higher net Sharpe than either component alone.

3.4 Walk-forward design & refit window

I train on a rolling 12-month window and refit every month. One year of roughly 2,000 names per day gives ample data for stable coefficients while staying short enough to track regime drift, and monthly refitting keeps the model current. Shorter windows made the coefficients noisier and longer ones adapted too slowly, so I fixed 12 months with monthly refits in advance.

I set weekly rebalancing in advance, and then confirmed it rather than tuned to it: as Section 4.4 shows, the net Sharpe is essentially flat across daily, weekly, and monthly rebalancing (1.35, 1.41, 1.29), so the choice is robust while trading far less than daily.

I also shaped the feature set. I dropped the industry one-hot dummies, which encode static industry means that do not predict the demeaned forward return and caused ill-conditioning; industry and factor exposure are handled instead by neutralization. And I tested classic price and volume features (short-term reversal, momentum, realized volatility) but found that at this two-day horizon they hurt out-of-sample performance, adding turnover with no return, so I rejected them.

4 Portfolio Construction and Results

4.1 From alpha to a dollar-neutral book

Each day we demean the smoothed forecast across the cross-section, so the book is dollar neutral by construction, then scale the positive side to sum to plus one and the negative side to minus one. Weights are proportional to forecast conviction. Because the alpha is spread over roughly 2,000 names, single-name weights are naturally tiny, so no explicit position cap is needed. The raw daily forecast is noisy, so trading it directly flips positions day to day and runs up turnover and noise-driven PnL swings. I therefore smooth each name’s forecast over time with a causal (trailing-only) exponentially weighted moving average before forming weights, which makes positions persistent, strips noise, and sharply cuts turnover, raising the net Sharpe. I use a roughly one-month span (about 21 trading days) because it balances responsiveness against turnover: shorter spans barely cut turnover, longer ones lag the forecast and let it go stale. One month removes most of the turnover (roughly 0.48 to 0.07 on a representative book, with the Sharpe flat to slightly higher) while keeping the forecast fresh and matching the signals’ multi-week persistence. I then build a risk-factor-neutral variant by regressing the forecast on the six risk factors each day and trading the residual.

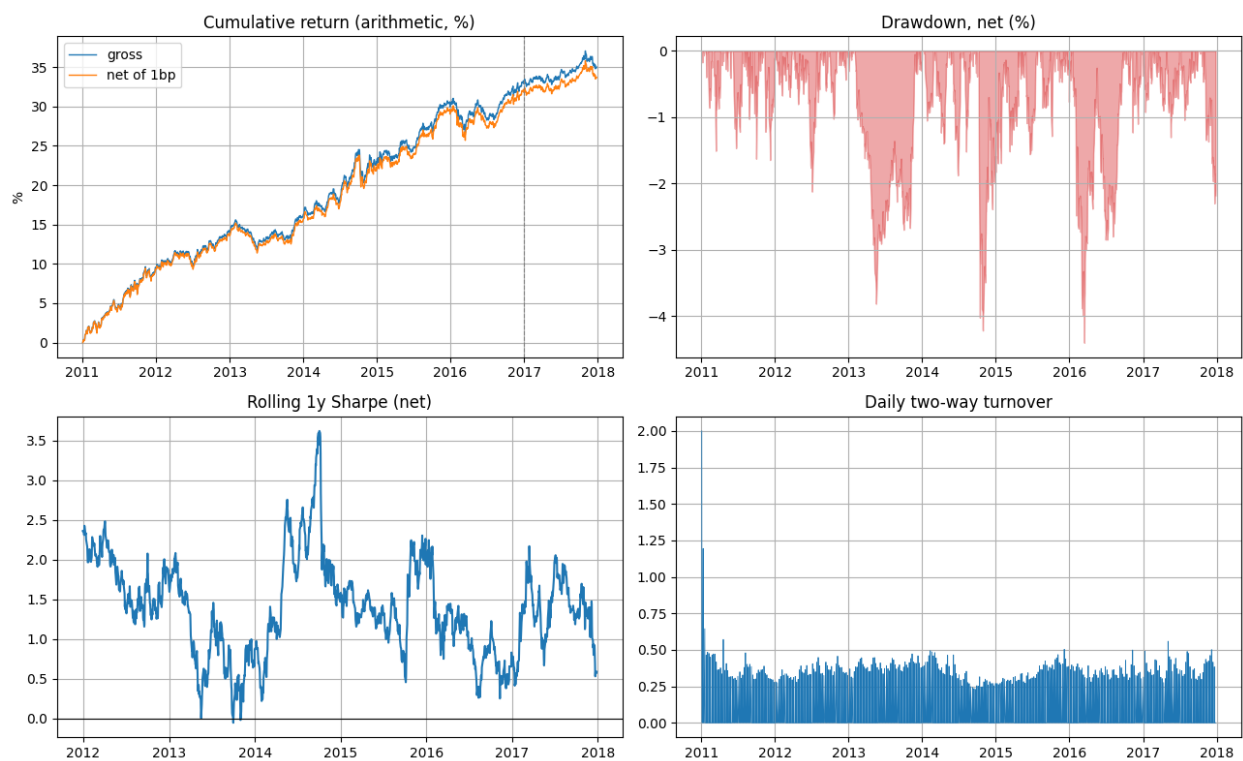
4.2 Performance of the final model

Cumulative return is the arithmetic running sum of daily returns, holding book size constant, as specified. Figure 5 shows the equity curve, the drawdown, the rolling one-year Sharpe ratio, and turnover for the final model. Table 2 lists the full statistics, net of 1 basis point per side, with weekly rebalancing.

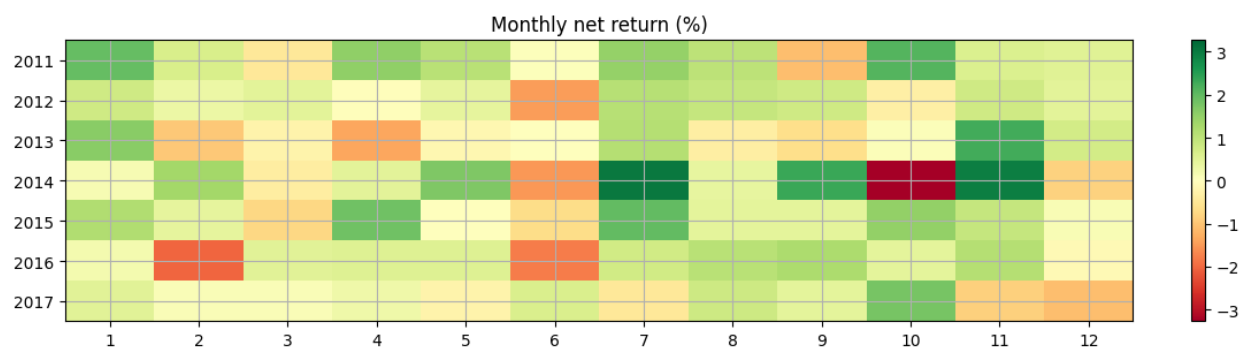
Table 2: Final model performance (ensemble, risk-factor-neutral, weekly, net of 1 bp).

Statistic	Full OOS (2011-2017)	2017 only
Annualized return	4.83%	1.94%
Annualized volatility	3.42%	2.88%
Net Sharpe ratio	1.41	0.67
Max drawdown (magnitude)	-4.40%	-2.30%
Max drawdown length	175 days	38 days
Positive days	55.3%	56.2%
Average daily turnover	0.073	0.076
Annual cost drag	0.19%	0.19%

The book delivers a net Sharpe of 1.41 at only 3.4% volatility, with a shallow worst drawdown of 4.4% that recovers within about eight months, and it is profitable on 55% of days. The volatility is deliberately low because the largest single improvement is risk-factor neutralization: as Table 3 shows, the same forecast traded without neutralization (the “base” variant) runs at 11.0% volatility with a Sharpe of only 0.36 and a 27.6% drawdown. Neutralizing the six factors cuts volatility from 11.0% to 3.4% and lifts the Sharpe from 0.36 to 1.41, which confirms the data-inspection finding that most of the base book’s risk is uncompensated factor exposure rather than alpha. The ensemble also beats each component: the IC-composite reaches 1.33 and the ridge model 1.26, so blending the two adds about 0.08 of Sharpe.



(a) Final model: cumulative return, drawdown, rolling 1y Sharpe, turnover



(b)

Figure 5

Table 3: Leaderboard of all books, ranked by net Sharpe (full OOS, weekly, net of 1 bp).

Book	IC	Net Sharpe	Ann. return	Ann. vol	Max DD	Turnover
ensemble, rf-neutral	0.0098	1.41	4.83%	3.42%	-4.40%	0.073
composite, rf-neutral	0.0097	1.33	4.56%	3.42%	-4.69%	0.058
ridge, rf-neutral	0.0081	1.26	4.53%	3.59%	-4.15%	0.127
composite, base	0.0097	0.92	6.65%	7.23%	-12.66%	0.048
ensemble, base	0.0098	0.36	3.96%	11.00%	-27.61%	0.053
ridge, base	0.0081	0.04	0.56%	14.07%	-40.59%	0.059

4.3 Impact of trading cost

Cost scales only with turnover, so we sweep the assumed cost against the same gross series (Table 4). The strategy has substantial cost headroom: each basis point of cost removes about 0.19% of annual return, the net Sharpe is still 1.20 at 5 bp and 0.92 at 10 bp, and the breakeven cost at which the net Sharpe reaches zero is roughly 26 bp per side. For a book that trades only about 7% of its notional per day, a realistic 1 to 2 bp assumption leaves the result essentially intact.

Table 4: Net Sharpe and annual return versus assumed trading cost.

Cost (bps, one-way)	0	1	2	3	5	7.5	10
Net Sharpe	1.47	1.41	1.36	1.30	1.20	1.06	0.92
Annual return	5.02%	4.83%	4.65%	4.46%	4.09%	3.63%	3.16%

4.4 Impact of rebalance frequency

We also vary how often the target weights are refreshed (Table 5). Weekly rebalancing is the sweet spot. Daily rebalancing chases noise and roughly doubles turnover (0.152 versus 0.073) for a slightly lower Sharpe, while monthly rebalancing halves turnover again but lets the forecast go stale, lowering the Sharpe to 1.29. Weekly keeps almost all of the signal at a fraction of the trading.

Table 5: Net Sharpe and turnover by rebalance frequency.

Rebalance	Net Sharpe	Ann. return	Turnover	Annual cost
Daily	1.35	4.64%	0.152	0.38%
Weekly	1.41	4.83%	0.073	0.19%
Monthly	1.29	4.41%	0.039	0.10%

5 Future Improvements and Concerns

Here are some potential improvements and next steps:

- **Recent-period strength.** The Sharpe over the full window is 1.41, but the 2017-only slice is 0.67 on a single year of data. This is a thin estimate, yet it is a reminder to monitor whether the edge is fading and to keep the walk-forward refit running.

- **Signal opacity and concentration.** The strongest signals (d1 to d7) are observed on only about 1.5% of name-days. The forecast leans on sparse, proprietary inputs, which is a concentration and data-availability risk.
- **Cost realism and capacity.** We model cost as linear in turnover. A real book would face market impact and borrow costs that grow with size, so the 3.4% volatility, while safe, also limits standalone return; scaling up would require a capacity and impact study.
- **Target Engineering** Currently a single 2-day-ahead return. We could try multi-horizon or denoised targets.
- **Dynamic signal reweighting** The rolling-IC plot shows IC drifts over time. The IC-composite already uses a trailing-window IC. I'd want to extend this to decay-weighted or shrinkage IC estimates, or weights conditioned on a regime variable (cross-sectional dispersion, volatility), could adapt faster than a flat 12-month window.